STATISTICS IN MEDICINE
Statist. Med. 2003; **22**:689–703 (DOI: 10.1002/sim.1362)

Sequential designs for phase III clinical trials incorporating treatment selection

Nigel Stallard*,† and Susan Todd

Medical & Pharmaceutical Statistics Research Unit, The University of Reading, P.O. Box 240, Earley Gate, Reading, Berkshire RG6 6FN, U.K.

SUMMARY

Most statistical methodology for phase III clinical trials focuses on the comparison of a single experimental treatment with a control. An increasing desire to reduce the time before regulatory approval of a new drug is sought has led to development of two-stage or sequential designs for trials that combine the definitive analysis associated with phase III with the treatment selection element of a phase II study. In this paper we consider a trial in which the most promising of a number of experimental treatments is selected at the first interim analysis. This considerably reduces the computational load associated with the construction of stopping boundaries compared to the approach proposed by Follman, Proschan and Geller (*Biometrics* 1994; **50**:325–336). The computational requirement does not exceed that for the sequential comparison of a single experimental treatment with a control. Existing methods are extended in two ways. First, the use of the efficient score as a test statistic makes the analysis of binary, normal or failure-time data, as well as adjustment for covariates or stratification straightforward. Second, the question of trial power is also considered, enabling the determination of sample size required to give specified power. Copyright © 2003 John Wiley & Sons, Ltd.

KEY WORDS: group sequential test; phase II/III trial; screening design; select and test design; spending functions

1. INTRODUCTION

The question of how to design a trial for the selection of one of a number of experimental therapies is not a new one. Early work included that by Paulson [1] and Bechhofer *et al.* [2].

More recently, Thall *et al.* [3, 4] and Schaid *et al.* [5] considered two-stage designs, with selection of the most promising from a number of treatments at the end of the first stage and comparison of this treatment with a control at the end of the second. In both cases, the final analysis is adjusted to allow for the selection procedure so that the overall type I error rate is maintained. Thall *et al.* consider the case of binary data and Schaid *et al.*

*Correspondence to: Nigel Stallard, Medical & Pharmaceutical Statistics Research Unit, The University of Reading, P.O. Box 240, Earley Gate, Reading, Berkshire RG6 6FN, U.K.
†E-mail: n.stallard@reading.ac.uk

Received November 2000
Accepted June 2002

the case of failure time response data. Follmann *et al.* [6] have also proposed an approach to the design of clinical trials in which an experimental treatment is selected at one of a number of interim analyses in a sequential design whilst preserving the overall type I error rate.

The work of this paper generalizes the designs of Thall *et al.* and Schaid *et al.* in two ways. First, by using the efficient score as a test statistic, our method is applicable for binary, normally distributed or failure time responses and allows the incorporation of covariate information at the interim and final analyses. Second, we consider a sequential trial in which a number of interim analyses comparing the selected and control treatments are performed. The work of Follman *et al.* is extended by providing a method by which the power at some specified alternative hypothesis can be achieved.

Although a restriction compared to the Follman *et al.* method is imposed, in that the selection of the most promising treatment must be made at the first interim analysis, this gives the advantage of a considerable reduction in the computational burden associated with the construction of critical values for the tests. The Follmann *et al.* approach requires evaluation of tail areas of multivariate normal distributions, the dimension of which increases with the number of treatments considered. In contrast, in the method proposed here, the dimensionality of the required integration is not beyond that needed in the single experimental treatment case.

The work described in this paper has been motivated by a number of trials about which we, or colleagues, have been consulted. A common aim of these trials has been to select the best of a small number of treatments and to provide a valid comparison of this with a control treatment whilst stopping recruitment to inferior treatments as quickly as possible. Even where the different treatments are different doses of the same drug, there has often been a reluctance to make any assumptions about the form of the dose–response relationship, so that a common approach for different doses and different treatments is appropriate. Traditionally, this aim might be met by conducting a phase II study in which the selection is made, followed by a phase III trial to compare this with control. The advantage of the methods described in the opening paragraphs and presented in this paper is that selection and comparison with control are conducted in the same study. In effect, the phase II patients are included in the phase III analysis, reducing the total resources required.

Although details of many of the motivating problems remain confidential, an example of a trial in which a number of different doses of galantamine are compared with a control for the treatment of Alzheimer's disease is used as an illustration.

In this paper we will present designs for sequential trials that compare k experimental treatments, T_i, $i=1,\ldots,k$, with a control treatment T_0 whilst maintaining the overall type I error rate. At the first interim analysis, the most promising experimental treatment is identified. If this treatment is sufficiently superior to the control treatment, or if it is sufficiently inferior to the control, the trial is stopped. If the trial is not stopped at the first interim analysis, additional patients are recruited to the selected treatment and control arms only and at least one further analysis is conducted.

The method for constructing the stopping limits is given in Section 2. The Alzheimer's example is described in Section 3 and is also used as the basis of a simulation study in Section 4. The paper concludes with a discussion in Section 5.

2. CONSTRUCTION OF SEQUENTIAL STOPPING RULES FOR A TEST INCLUDING TREATMENT SELECTION

2.1. Test statistics and notation

In a sequential trial to compare a single experimental therapy with control, a test statistic measuring the treatment difference is compared with stopping limits at a series of analyses. The trial stops once a stopping limit is exceeded. Otherwise, it continues up to some maximum number, n, of inspections.

Let θ be a measure of the superiority of the experimental therapy, with $\theta > 0$ corresponding to the experimental treatment being superior to the control, $\theta < 0$ corresponding to inferiority to control and $\theta = 0$ corresponding to control and experimental treatment being of equal efficacy. We wish to test the null hypothesis $H_0\colon \theta = 0$, with one-sided type I error rate $\alpha/2$, that is to conclude that $\theta > 0$ when in fact $\theta \leqslant 0$ with probability at most $\alpha/2$. Whitehead [7] proposes use of the efficient score for θ as a test statistic, which may be compared with stopping limits which depend on the observed Fisher's information at the inspection time. The efficient score and observed information at the jth interim analysis will be denoted by Z_j and V_j, respectively.

Forms for θ and the corresponding efficient score and observed information for a number of settings are given by Whitehead [7]. For example, in the case of binary data, θ might be taken to be the log-odds ratio, $\log((p_E(1-p_C))/((1-p_E)p_C))$, where p_E and p_C are the probabilities of success in experimental and control groups, respectively. If the experimental and control groups have, respectively, n_E and n_C observations, with the estimated success rates $\widehat{p_E}$ and $\widehat{p_C}$, Z and V are given, respectively, by $Z = n_E n_C(\widehat{p_E} - \widehat{p_C})/n$ and $V = n_E n_C(n_E\widehat{p_E} + n_C\widehat{p_C})(n_E(1 - \widehat{p_E}) + n_C(1 - \widehat{p_C}))/n^2$ where $n = n_C + n_E$. In the case of normal data, θ might be taken to be the standardized difference in means. If the experimental and control groups have observed means $\bar{x}_E$ and $\bar{x}_C$, respectively, and the sum of the squares of all observations (taken over both groups) is Q, then Z and V are given by $Z = n_E n_C(\bar{x}_E - \bar{x}_C)/nD$ and $V = n_E n_C/n - Z^2/2n$, where $D = \sqrt{(Q/n - (n_E\bar{x}_E + n_C\bar{x}_C)^2/n^2)}$. In this case, under the null hypothesis, when the expected value of Z is zero, if $n_E = n_C$, V is approximately equal to $n_C/2$. Asymptotically, for small θ, $Z_1 \sim \mathrm{N}\,(\theta V_1, V_1)$ and $Z_j - Z_{j-1} \sim \mathrm{N}\,(\theta(V_j - V_{j-1}), V_j - V_{j-1})$, with $Z_j - Z_{j-1}$ independent of Z_{j-1} for $j = 2,\ldots,n$. Stopping rules for $Z_1,\ldots,Z_n$ for the comparison of two treatments can be obtained based on this asymptotic distribution via the spending function approach [8, 9], as described in detail in Section 2.2.

In the case where there are k experimental treatments, let θ_i be a measure of the superiority of T_i over T_0 for $i = 1,\ldots,k$, with $\theta_i > 0, \theta_i = 0$ and $\theta_i < 0$ corresponding to the states T_i superior to, of equal efficacy as, and inferior to, T_0. In this case a false positive result can be obtained in several different ways. We wish the probability of selecting treatment T_i and concluding that $\theta_i > 0$ when in fact $\theta_i \leqslant 0$ to be at most $\alpha/2$. This is achieved by specifying that the probability of concluding any θ_i exceeds 0 under the null hypothesis $H_0\colon \theta_1 = \cdots = \theta_k = 0$ is at most $\alpha/2$.

For $i = 1,\ldots,k$, let $Z_{i,1}$ and $V_{i,1}$, respectively, denote the efficient score and observed Fisher's information for θ_i, at the first interim analysis. Assuming common $V_{i,1}$, the experimental treatment with the greatest estimated efficacy will be that with the largest observed $Z_{i,1}$. This treatment will be selected for further testing and all other experimental treatments will be dropped after the first interim analysis. Let the selected treatment be denoted by T_S so that S is a discrete random variable taking values in $\{1,\ldots,k\}$ indicating the selected treatment and $Z_{S,1} = \max_{i=i,\dots,k}\{Z_{i,1}\}$. Since S has been chosen precisely because of the large value of $Z_{S,1}$ observed, the distribution of $Z_{S,1}$ will not be normal, so that adjustment for the treatment selection is necessary in subsequent interim analyses. The distribution of $Z_{S,1}$ is obtained in Section 2.3, and a method for the construction of stopping boundaries based on this distribution is given in Section 2.4.

2.2. Construction of sequential stopping rules for a single treatment comparison using the spending function approach

Returning to the case of a single treatment comparison, suppose that a stopping rule is imposed such that, for $j = 1, \dots, n-1$, the test continues to the $(j+1)$th interim analysis if the jth interim analysis takes place and $Z_j \in (l_j, u_j)$.

A stopping rule with specified properties can be constructed via the spending function approach due to Lan and DeMets [8]. This approach was extended by Kim and DeMets [9] to give a test of the null hypothesis against a two-sided alternative. These tests are symmetric, with $l_j = -u_j$. A test of the one-sided alternative with both upper and lower stopping limits, in which crossing of the lower boundary leads to stopping of the study without rejection of the null hypothesis, can also be constructed via the spending function approach. Two different spending functions are used to give the upper and lower boundaries. The upper boundary is defined by $\alpha^*_{\mathrm{U}}: [0,1] \to [0, \alpha/2]$, a non-decreasing function with $\alpha^*_{\mathrm{U}}(0) = 0$, $\alpha^*_{\mathrm{U}}(1) = \alpha/2$, by

$$\mathrm{pr}(Z_j \geqslant u_j, Z_1 \in (l_1, u_1), \dots, Z_{j-1} \in (l_{j-1}, u_{j-1}) | H_0) = \alpha^*_{\mathrm{U}}(t_j) - \alpha^*_{\mathrm{U}}(t_{j-1}) \quad (1)$$

for $j = 1, \dots, n$, where t_j is the proportion of planned maximum information available at the jth interim analysis with t_0 taken to be 0. The lower boundary may be defined in one of two ways. Pampallona *et al.* [10] described how a β spending function, $\beta^*: [0,1] \to [0, \beta]$, a non-decreasing function with $\beta^*(0) = 0$ and $\beta^*(1) = \beta$ may be used where

$$\mathrm{pr}(Z_j \leqslant l_j, Z_1 \in (l_1, u_1), \dots, Z_{j-1} \in (l_{j-1}, u_{j-1}) | H_1) = \beta^*(t_j) - \beta^*(t_{j-1})$$

Alternatively, Stallard and Facey [11] define the lower boundary using a non-decreasing function $\alpha^*_{\mathrm{L}}: [0,1] \to [0, 1-\alpha/2]$, with $\alpha^*_{\mathrm{L}}(0) = 0$ and $\alpha^*_{\mathrm{L}}(1) = 1 - \alpha/2$. This might properly be called a '$1 - \alpha/2$ spending function'. The lower boundary is then given by the expression

$$\mathrm{pr}(Z_j \leqslant l_j, Z_1 \in (l_1, u_1), \dots, Z_{j-1} \in (l_{j-1}, u_{j-1}) | H_0) = \alpha^*_{\mathrm{L}}(t_j) - \alpha^*_{\mathrm{L}}(t_{j-1}) \quad (2)$$

With the method of Pampallona *et al.*, the power of the test is guaranteed, but a numerical search over $V_{\max}$ with fixed $t_1, \dots, t_n$ is required to ensure that $l_n = u_n$. With the Stallard and Facey method, the boundaries meet, but a search is required to give the desired power. The method due to Stallard and Facey will be used below. It is, however, easy to see how the Pampallona *et al.* approach might be applied in a similar fashion.

Recursive numerical integration can be used to evaluate the probabilities in (1) and (2). This approach was first described by Armitage *et al.* [12], and has been described by authors such as Jennison [13].

Conditional on the observed value of V_1, the density of Z_1 is given by

$$f_1(z;\theta)=\frac{1}{\sqrt{V_1}}\,\phi\left(\frac{z-\theta V_1}{\sqrt{V_1}}\right) \tag{3}$$

so that

$$\mathrm{pr}(Z_1 \geqslant u_1;\theta)=\Phi\left(\frac{u_1-\theta V_1}{\sqrt{V_1}}\right) \tag{4}$$

For $j=2,\ldots,n$, the conditional distribution of Z_j given $Z_1=z_1,\ldots,Z_{j-1}=z_{j-1}$ depends only on Z_{j-1} and is given by

$$f_j(z_j\,|\,z_{j-1};\theta)=\frac{1}{\sqrt{(V_j-V_{j-1})}}\,\phi\left(\frac{z_j-z_{j-1}-\theta(V_j-V_{j-1})}{\sqrt{(V_j-V_{j-1})}}\right) \tag{5}$$

and

$$\begin{aligned}&\mathrm{pr}(Z_j \geqslant u_j, Z_1\in(l_1,u_1),\ldots,Z_{j-1}\in(l_{j-1},u_{j-1});\theta)\\ &\quad=\int_{l_1}^{u_1}\cdots\int_{l_{j-1}}^{u_{j-1}}\int_{u_j}^{\infty} f_1(z_1;\theta)f_2(z_2\,|\,z_1,\theta)\cdots f_j(z\,|\,z_{j-1},\theta)\,\mathrm{d}z_j\ldots\mathrm{d}z_1\end{aligned} \tag{6}$$

with probabilities that $Z_j \leqslant l_j$ for $j=1,\ldots,n$ given similarly.

Numerical calculation of the integrals given in (4) and (6) can be used in a search procedure to find $\{(l_j,u_j):\ j=1,\ldots,n\}$ to satisfy (1) and (2).

2.3. Distribution of the test statistic for a first interim analysis incorporating treatment selection

In the k-comparison case, for small θ_i, each $Z_{i,1}$ has an asymptotic $\mathrm{N}(\theta_i V_1, V_1)$ distribution, as stated in Section 2.1. The inclusion of a common control leads to correlation between $Z_{i,1}$ and $Z_{j,1}$ for $i\neq j$.

If equal numbers of observations are taken on each treatment, under the null hypothesis that all treatments are equally efficacious, the expected values of $V_{i,1}, i=1,\ldots,k$ are equal. If it is assumed that equal information is available on all groups, so that in particular, $V_{1,1}=\cdots=V_{k,1}$, with this common value denoted V_1, then for $i\neq j$, $\mathrm{cov}(Z_{i,1},Z_{j,1})=V_1/2$ (see Appendix). The effects of this assumption will be assessed via simulation studies in Section 4. Asymptotically for large sample sizes and small $\theta_1,\ldots,\theta_k$

$$\begin{pmatrix}Z_{1,1}\\ Z_{2,1}\\ \vdots\\ Z_{k,1}\end{pmatrix}\sim \mathrm{MN}\left(\begin{pmatrix}\theta_1 V_1\\ \theta_2 V_1\\ \vdots\\ \theta_k V_1\end{pmatrix},\begin{pmatrix}V_1 & V_1/2 & \cdots & V_1/2\\ V_1/2 & V_1 & \cdots & V_1/2\\ \vdots & \vdots & \ddots & \vdots\\ V_1/2 & V_1/2 & \cdots & V_1\end{pmatrix}\right) \tag{7}$$

It is shown in the Appendix that the joint density of $Z_{S,1}$ and S, where S is the random variable giving the selected treatment, is given by

$$f_{Z_{S,1},S}(z_1,i)=\int_{-\infty}^{\infty}\frac{1}{\surd(V_1/2)}\,\phi\left(\frac{x-z_1}{\surd(V_1/2)}\right)f_{X_S,S}(x,i)\,\mathrm{d}x \tag{8}$$

where

$$f_{X_S,S}(x,i)=\frac{1}{\surd(V_1/2)}\,\phi\left(\frac{x-\theta_i V_1}{\surd(V_1/2)}\right)\prod_{i'\neq i}\Phi\left(\frac{x-\theta_{i'}V_1}{\surd(V_1/2)}\right)$$

The probability that $Z_{S,1}$ exceeds u_1 and $S=i$ is shown in the Appendix to be equal to

$$\int_{-\infty}^{\infty}f_{X_S,S}(x,i)\left[1-\Phi\left(\frac{u_1-x}{\surd(V_1/2)}\right)\right]\mathrm{d}x$$

The marginal distribution of $Z_{S,1}$ can be obtained by summing over $i=1,\ldots,k$.

Let $Z_{S,j}$ and $V_{S,j}$ denote, respectively, the efficient score and observed information for θ_S at the jth interim analysis, $j=2,\ldots,n$, following the selection of treatment S. Since only treatments T_S and T_0 are used at the second and subsequent inspections, the first subscript for V will be omitted.

Assuming the trial has not stopped prior to the jth inspection, the change in Z, $Z_{S,j}-Z_{S,j-1}$, for $j=2,\ldots,n$, is asymptotically normally distributed with mean $\theta_S(V_j-V_{j-1})$ and variance (V_j-V_{j-1}), so that the conditional distribution of $Z_{S,j}$ given $Z_{S,j-1}=z_{j-1}$ is still given by (5) with θ set equal to θ_S.

2.4. Construction of sequential stopping rules in the treatment selection case

In addition to the specification of null properties, the stopping rule is required to have specified power. The power of a selection study is discussed by Thall *et al.* [3,4]. In their approach, constants δ_1 and δ_2 are specified with $\delta_2>\delta_1\geqslant 0$ such that $\theta_i=\delta_1$ represents a marginal improvement of treatment T_i over the control and $\theta_i=\delta_2$ represents a clinically relevant improvement of T_i over the control. In the event that some treatment is superior to the control, that is for some $i,\theta_i\geqslant\delta_2$, and that no θ_i is in the interval (δ_1,δ_2), it is required that a treatment with $\theta_i\geqslant\delta_2$ is selected at the first stage and that the null hypothesis is rejected at the first or some subsequent stage with at least specified probability, which will be referred to as the power of the selection design.

Suppose that $\theta_1\geqslant\delta_2$ and that no $\theta_i, i=2,\ldots,k$, is in (δ_1,δ_2). The power of the design depends on the values of $\theta_2,\ldots,\theta_k$. Thall *et al.* showed that the power is at least $1-\beta$ for all values of $\theta_2,\ldots,\theta_k$ not in (δ_1,δ_2) if it is equal to $1-\beta$ in the case that they call the 'least favourable configuration', when $\theta_2=\cdots=\theta_k=\delta_1$, so that power should be evaluated taking this configuration as the alternative hypothesis H_1.

A sequential stopping boundary in which k experimental treatments are compared with a control treatment at the first interim analysis can be constructed so as to satisfy the error spending function equations (1) and (2) and to have overall power of $1-\beta$ at the alternative hypothesis $\theta_1=\delta_2,\theta_2=\cdots=\theta_k=\delta_1$ as follows.

Suppose a maximum of n interim analyses are planned, with the proportion of information available at the jth equal to t_j. Since the test statistics $Z_{i,1}$ and $Z_{S,j+1}-Z_{S,j}$ are not standardized, but have variance V_1 and $V_{j+1}-V_j$, the critical values for the test depend on the value of V_n in addition to $n, t_1, \ldots, t_n$ and α_U^* and α_L^*. We shall make this dependence explicit here and write $l_j(V_n)$ and $u_j(V_n)$. Critical values for the test can be found to satisfy (1) and (2) for $V_n = 1$, so that $V_j = t_j$ for $j = 1, \ldots, n$. At the first interim analysis, $u_1(1)$ and $l_1(1)$ are found to satisfy (1) and (2) using a numerical search based on the distribution of the test statistic Z_1 given by (8). For subsequent looks, the probability of crossing the upper and lower boundaries are given by (6) with $f_1(z_1)$ given by (8), so that $(l_j(1), u_j(1))$, $j = 2, \ldots, n$ can be found recursively to satisfy (1) and (2). A test with the same $n, t_1, \ldots, t_n$ and α_U^* and α_L^* with $V_n = V_{\max}$ so that $V_j = t_j V_{\max}, j = 1, \ldots, n$, will satisfy (1) and (2) if it has critical values $u_j(V_{\max}) = u_j(1)\sqrt{V_{\max}}$ and $l_j(V_{\max}) = l_j(1)\sqrt{V_{\max}}$ for $j = 1, \ldots, n$. Stopping boundaries can therefore be obtained for any value of V_n without repeating the recursive numerical integration. The power of a test is evaluated by calculating the probability of crossing the upper boundary under the alternative hypothesis using (8) and (6), so that a numerical search can be used to find V_n to give required power.

A program written in C to implement this method is available from the authors. On a Sun Ultra 5 workstation, the calculations to give the design given in Table IV took 20 seconds.

3. EXAMPLE: TREATMENT SELECTION IN A STUDY OF ALZHEIMER'S DISEASE

In this section, the approach described in Section 2 is illustrated by showing how it could have been implemented in a phase III clinical trial to assess a new treatment for the alleviation of Alzheimer's disease. The trial, which was conducted by Shire Pharmaceutical Development Limited, is described in detail by Wilkinson and Murray [14]. Eligible patients with mild or moderate Alzheimer's disease were randomized to one of four treatment groups. These were placebo and the drug under investigation, galantamine, at three dose levels, 18 mg per day, 24 mg per day and 36 mg per day for twelve weeks. The primary endpoint was assessment according to the Alzheimer's Disease Assessment Scale (ADAS) cognitive portion, which was assumed to be normally distributed.

The study was originally designed as a sequential trial with the first interim analysis conducted when data were available from 20 patients on each treatment arm and subsequent analyses conducted on 40 patients per treatment, 60 patients per treatment and so on. The comparisons of the three doses of galantamine with placebo were monitored using three separate triangular test stopping boundaries (reference [7], Section 3.7), with no allowance made for the use of more than one experimental treatment. It was planned that recruitment to a treatment arm be terminated either if the upper boundary of the test comparing that arm with the placebo was crossed for superiority of that dose over control or if the lower boundary of the test comparing the arm to placebo was crossed for futility. Recruitment to the placebo arm continued as long as recruitment to any other arm continued. A maximum of 100 patients on any treatment arm was anticipated.

A summary of data comprising Z and V for comparison of each dose group with the placebo group at the first interim analysis is given in Table I. It can be seen that at the first interim analysis, the largest value of Z is for the 24 mg group. The Z values for no groups were outside the triangular test continuation region at this look so recruitment to all four treatment groups continued to the second interim analysis. The values of Z and V for the comparison of the different dose groups with the placebo group at the second interim analysis are also given

Table I. A summary of relevant data from the galantamine study.

	Analysis 1	Analysis 2
(Z, V) (18 mg/day versus placebo)	$(4.28, 11.79)$	$(4.29, 20.61)$
(Z, V) (24 mg/day versus placebo)	$(7.89, 10.82)$	$(13.78, 19.58)$
(Z, V) (36 mg/day versus placebo)	$(4.64, 11.01)$	$(8.39, 20.30)$

Table II. Spending function, boundary values and error rates for a boundary obtained allowing for treatment selection.

	Analysis				
	1	2	3	4	5
α_U^*	0.001	0.010	0.019	0.024	0.025
α_L^*	0.185	0.733	0.933	0.973	0.975
Upper boundary values	10.6	12.0	13.5	15.1	16.8
Lower boundary values	−0.6	4.5	9.6	14.5	16.8
Pr(cross upper boundary; H_0)	0.001	0.010	0.019	0.024	0.025
Pr(cross lower boundary; H_0)	0.185	0.733	0.933	0.973	0.975

in Table I. The value of Z for the 24 mg group of 13.78 led to crossing of the upper boundary and recruitment to this arm was terminated. Recruitment to the 36 mg arm continued until the arm was dropped due to safety concerns at the second analysis. Recruitment to the 18 mg and placebo arms continued until the lower boundary was crossed at the fourth interim analysis. It was then concluded that the 24 mg/day dose was safe and effective, although another trial [15] indicated that a lower dose was also effective.

A sequential design allowing for the treatment selection has been obtained using the method described in Section 2. In the notation of Section 2, $k=3$ and $n=5$. As shown in Section 2.1, the expected value of V at each interim analysis is equal to half the number of patients on each arm, so that $V_1, \ldots, V_5$ may be taken to be 10, 20, 30, 40 and 50. The spending function corresponding to the triangular test was used to give a test that would stop early under the null hypothesis in a similar way to the design that was actually used. For five equally spaced inspections, the values of $\alpha_U^*(t)$ and $\alpha_L^*(t)$ for the triangular test with one-sided type I error rate 0.025 are given in the first two rows of Table II.

Boundary values for the test designed using the approach of Section 2 are given in the third and fourth rows of Table II. The cumulative probability under the null hypothesis of crossing the upper and lower boundary at or before each interim analysis for this test are given in the fifth and sixth rows. It can be seen that the boundary crossing probabilities are exactly equal to the spending functions as required.

Using this design, recruitment to the 18 mg and 36 mg groups would have ceased at the first interim analysis with only the 24 mg and placebo groups continuing. The trial would then have stopped at the second look and the conclusion been drawn that the 24 mg dose is effective.

It is interesting to compare the boundary given in Table II with the spending function boundary that would be obtained if the treatment selection was ignored, that is, if k was

Table III. Boundary values and error rates for a boundary obtained ignoring treatment selection.

	Analysis				
	1	2	3	4	5
Upper boundary values	9.6	10.6	12.0	13.6	15.2
Lower boundary values	−2.8	2.7	8.1	12.9	15.2
Pr(cross upper boundary; H_0)	0.003	0.022	0.038	0.048	0.049
Pr(cross lower boundary; H_0)	0.044	0.582	0.882	0.948	0.951

Table IV. Boundary values for test with power 0.9 to select T_1 when $\theta_1 = 0.55$ and $\theta_2 = \theta_3 = 0$.

	Analysis				
	1	2	3	4	5
V	13.6	27.2	40.7	54.3	67.9
Upper boundary values	12.3	13.9	15.8	17.6	19.5
Lower boundary values	−0.7	5.2	11.2	16.9	19.5

taken to be 1. Boundary values for this test are given in the first two rows of Table III. Comparing Tables II and III shows that allowing for the treatment selection results in both upper and lower boundaries being moved upwards relative to the incorrect design to allow for the expected increase in Z arising from the treatment selection. The null probabilities of crossing the upper and lower boundary when this test is used for the trial with $k = 3$ are given in the third and fourth rows of Table III. As expected, even under the null hypothesis that all treatments are identical to the control, if the best is selected the expected value of Z is larger than zero and there is a greater chance of crossing the upper boundary than planned. In this case the probability of crossing the upper boundary is 0.049, nearly twice the nominal one-sided error rate, illustrating the need for a valid sequential design when the best among even a small number of treatments is selected.

Suppose we desire power of 0.9 to select dose i and conclude that it is superior to the control when $\theta_i = 0.55$ and $\theta_j = 0$ for all $j \neq i$, where θ_i is the standardized difference between dose i and control, that is the power at the alternative hypothesis that $\delta_2 = 0.55$ and $\delta_1 = 0$ in the notation of Section 2.4. This is a similar requirement to that of the actual trial, in which each comparison of a treatment arm with control had power of 0.9 to detect an improvement corresponding to $\theta = 0.55$. With five equally spaced interim analyses as planned, this power is achieved if V_5 is equal to 67.9, corresponding to inclusion of 27 patients on each treatment arm for each interim analysis. The boundary values for this test are given in Table IV. It can be seen that the critical values for the test in Table IV are given by multiplying those in Table II by $\sqrt{(67.9/50)}$.

Of additional interest is the timing of the first interim analysis. An early analysis gives little information on which to base the selection of the most promising treatment, whilst a

Table V. Expected number of patients required to give power of 0.9 for a range of values of V_1.

	V_1/V_5								
	0.1	0.12	0.14	0.16	0.18	0.2	0.4	0.6	0.8
$E(n\,\vert\,H_0)$	193.6	180.5	173.7	170.3	169.2	169.5	219.8	299.7	346.3
$E(n\,\vert\,H_1)$	190.2	184.0	181.2	180.5	181.3	183.0	230.1	301.9	346.3

late analysis requires a large number of patients to be included on all treatment arms, and so is wasteful of resources.

To investigate this question, a number of designs were obtained. In each case five interim analyses were planned, the timing of the first being varied and the remaining four equally spaced. The value of V_5 was obtained in each case to give power 0.9 to conclude that treatment T_1 was the most promising and was superior to the control under the alternative hypothesis that $(\theta_1, \theta_2, \theta_3) = (0.55, 0, 0)$. Using the approximation that V is twice the number of patients in each group, the expected number of patients required under null and alternative hypotheses is given in Table V for V_1 varying between $0.1V_5$ and $0.8V_5$. It can be seen that in this case, the minimum expected number patients required to satisfy the power requirement is achieved under the null when $V_1 = 0.18V_5$ and under the alternative when $V_1 = 0.16V_5$. Although this aspect of the design was not considered, the choice of the first interim analysis at one-fifth of V_5 as actually conducted, actually gives an expected number of patients close to the minimum.

4. A SIMULATION STUDY

The properties of the test given in Table II are evaluated based on the assumption that Fisher's information for the estimation of the mean response in all groups at the first interim analysis is equal, so that in particular, $V_{1,1} = \cdots = V_{k,1}$. This is known to hold asymptotically under the null hypothesis if groups are of equal size. In small samples, or in the case when the numbers of observations on the different arms differ, however, the assumption that $V_{1,1} = \cdots = V_{k,1}$ may be violated. In order to assess how violations of this assumption might affect the properties of a real trial, a simulation study was conducted based on the example just considered. The triangular test boundary given in Table IV was used. Two simulation studies were conducted. Simulations were conducted with the planned 27 patients per treatment arm recruited for each interim analysis and with the number of patients in each group deviating from 27 by a positive or negative (each with probability one-half) Poisson random variable with mean 2 to represent small differences between group sizes and means 5 and 10 to investigate the effect of larger differences. To estimate the type I error rate and power, the data were simulated under the null and alternative hypotheses. Since θ_i is the standardized difference between the means, the type I error rate and power do not depend on the common variance used for the simulations, so that data from the control group were simulated from a normal distribution with mean 0 and variance 1. Under the null hypothesis, data from all treatment groups were also simulated

Table VI. Simulated error rates for the boundary given in Table IV.

	Cumulative null stopping probability Analysis					Power
	1	2	3	4	5	
27 patients per group						
Upper boundary values	0.001	0.008	0.018	0.024	0.024	0.885
Lower boundary values	0.186	0.732	0.934	0.974	0.976	
27 patients per group ± Poisson with mean 2						
Upper boundary values	0.001	0.009	0.017	0.023	0.024	0.890
Lower boundary values	0.187	0.732	0.935	0.975	0.976	
27 patients per group ± Poisson with mean 5						
Upper boundary values	0.001	0.009	0.018	0.023	0.024	0.873
Lower boundary values	0.185	0.736	0.935	0.975	0.976	
27 patients per group ± Poisson with mean 10						
Upper boundary values	0.001	0.009	0.018	0.023	0.024	0.818
Lower boundary values	0.185	0.741	0.938	0.975	0.976	

from this distribution, whilst under the alternative hypothesis the mean for one group was taken to be 0.55.

Estimated cumulative probabilities under the null hypothesis of stopping by the jth interim analysis for $j=1,\ldots,5$ on the upper or lower boundary are given in Table VI in each of the three cases considered together with estimated power probabilities, where this is the probability of selecting the truly superior treatment and crossing the upper boundary. In each case the estimates are based on 100 000 simulated trials, so that the standard errors of the estimated power and overall type I error rate are approximately 0.001 and 0.0005, respectively. The values given in Table VI suggest that the power may be reduced when there is large variation in group sizes, but that the type I error rate is generally preserved.

5. DISCUSSION

The question of how to design a clinical trial to select the best of a number of experimental therapies has attracted interest for many years. Recent reviews are given by Hochberg and Tamhane [16] and Bechhofer *et al.* [17]. In most of the methods proposed for treatment selection, however, a large sample size is required if the same trial is to lead both to selection of the most promising treatment and to comparison of this treatment with a control in which the type I error rate is fixed.

In practice, treatment selection is usually separated from confirmatory comparison with a control treatment, the selection being made in more informal phase II studies rather than in a more rigorous phase III clinical trial. In this paper a method is proposed that enables the process to be accelerated by combining the two studies, so that the 'phase II' patients can be included in the 'phase III' analysis. A sequential approach is taken in which the most promising treatment is selected at the first interim analysis.

The designs obtained here include, as special cases, both the two-stage 'select and test' designs of Thall *et al.* and the sequential trials for comparison of a single treatment with control described by, for example, Whitehead [7] and used in a number of clinical trials. Although a clear generalization of the designs obtained by these authors, the problem considered here is by no means the most general. In particular, it is required that at most one experimental treatment is selected at the first interim analysis, with no possibility of continuing with either a fixed number above one, or a random number of treatments. This means that the selection is necessarily based on a small amount of data. The proposed approach is appropriate in cases when one treatment is clearly superior to all others and it is desired to select this treatment and compare it with the control, or when a group of treatments are superior to the others and selection of any one of these is acceptable. The latter case is that considered by Thall *et al.* in their construction of the alternative hypothesis described in Section 2.4. If there is a group of treatments which are superior to the control and to all others and it is desired to select the best from this group, the approach described in this paper will not be appropriate, since it would be desirable to drop ineffective treatments early in the trial, but continue with more than one experimental treatment in addition to the control so as to have more data to determine which treatment is the best. Such an approach might require considerably larger sample sizes than those considered here.

Follman *et al.* [6] have also described a method for the construction of error-rate preserving sequential designs for studies with treatment selection. They allow two or more experimental treatments to be included in the second and subsequent interim analyses. Their method requires evaluation of tail areas of multivariate normal distributions, and so is computationally intensive, particularly when the number, k, of experimental therapies considered is large. This is also the approach used in the bivariate endpoint setting by Todd [18, 19] and for treatment selection by Vincent (University of Reading PhD thesis, 1998). Follmann *et al.* replace the numerical integration by Monte Carlo simulations to obtain approximate stopping limits. Our approach, in contrast, does not require integration of dimensionality above that required in the single experimental treatment case, so that computational requirements do not increase with k. More research is needed in this area to provide designs which allow treatments to be dropped at any interim analysis, and to assess when these designs are preferable to that developed here.

Further research would also be appropriate to address a number of other questions. The timing of the first interim analysis is considered in the example of Section 3, when it is shown that taking this first look when V is slightly less than one-fifth of its maximum value appears to be appropriate. The effect of the number of treatments to be compared, the number of subsequent interim analyses, and even the required power of the study on the optimal timing of this first interim analysis could all be investigated and would be of interest in practical application.

In practice, in order to accelerate the drug development programme, phase II trials sometimes use a rapidly observable response which is different from the response on which the primary phase III assessment is based. The decision of when to continue with phase III investigation after a phase II study is also determined by data on treatment safety as well as efficacy, with the most efficacious treatment dropped if it leads to safety concerns. The incorporation of these aspects into the selection designs proposed is the subject of continuing research.

Often in a trial of the type considered, the different treatments will be doses of the same drug. Sometimes it is appropriate to consider these as if they are different treatments and the approach of this paper will be most appropriate. In other cases some assumptions about the dose–response relationship might be made, so that data gained at one dose provide information on the likely response at others. Again, more research is needed to obtain error-rate preserving designs that use this additional information.

APPENDIX: DERIVATION OF DENSITIES AND STOPPING PROBABILITIES

The parameter θ_i, giving the difference between treatment T_i and the control treatment T_0, may be expressed as a difference $\mu_i - \mu_0$, where μ_i, $i = 0, \ldots, k$ is a measure of the level of response in group i. The efficient score for $\theta_i, Z_{i,1}$, can then be expressed in the form $V_{i,1}(\hat{\mu}_i - \hat{\mu}_0)$ where $\hat{\mu}_i$ is the maximum likelihood estimate of μ_i (see, for example, Jennison and Turnbull [20] and Scharfstein *et al.* [21]).

The statistics $\hat{\mu}_0, \ldots, \hat{\mu}_k$ are independent and, for large groups, by the central limit theorem approximately normally distributed. Since $V_{i,1}$ is the asymptotic null variance of $Z_{i,1}$, we must have $\mathrm{var}(\hat{\mu}_i) + \mathrm{var}(\hat{\mu}_0) = V_{i,1}^{-1}$. If we assume that equal information is available for each group, $\mathrm{var}(\hat{\mu}_0) = \cdots = \mathrm{var}(\hat{\mu}_k)$. Hence $V_{i,1}$ is the same for all i, and so may be written as V_1, so that asymptotically $\hat{\mu}_1 \sim \mathrm{N}(\mu_i, V_1^{-1}/2)$.

Let X_i denote the random variable $V_1(\hat{\mu}_i - \mu_0)$, $i = 0, \ldots, k$, then

$$\begin{pmatrix} X_0 \\ X_1 \\ \vdots \\ X_k \end{pmatrix} \sim \mathrm{MN}\left(\begin{pmatrix} 0 \\ \theta_1 V_1 \\ \vdots \\ \theta_k V_1 \end{pmatrix}, \begin{pmatrix} V_1/2 & 0 & \ldots & 0 \\ 0 & V_1/2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & V_1/2 \end{pmatrix} \right)$$

Since $Z_i = X_i - X_0$, Z_i has an asymptotic $\mathrm{N}(\theta_i V_1, V_1)$ distribution, with $\mathrm{cov}(Z_i, Z_j) = V_1/2$ for $i \neq j$, as stated in (6) in Section 2.1.

Since the treatment T_S with $Z_{S,1} = \max_{i=1,\ldots,k}\{Z_{i,1}\}$ will be selected at the first interim analysis, with this value of $Z_{S,1}$ compared with the boundary values, we are interested in the joint density of S and $Z_{S,1}$ and in the probability that $S = i$ and that $Z_{S,1}$ exceeds some value c.

The joint density of X_S and S, $f_{X_S,S}(x, i)$ is equal to $f_{X_i}(x)\mathrm{pr}(S = i \mid X_i = x)$. If $X_i = x$, then $S = i$ if $\max_{i' \neq i}\{X_{i'}\} < x$. As $X_1, \ldots, X_k$ are independent, that is with probability

$$\prod_{i' \neq i} \Phi\left(\frac{x - \theta_{i'}^{(1)} V_1}{\sqrt{(V_1/2)}} \right)$$

so that

$$f_{X_S,S}(x, i) = \frac{1}{\sqrt{(V_1/2)}} \phi\left(\frac{x - \theta_i V_1}{\sqrt{(V_1/2)}} \right) \prod_{i' \neq i} \Phi\left(\frac{x - \theta_{i'} V_1}{\sqrt{(V_1/2)}} \right)$$

As $Z_i = X_i - X_0$, the joint density of $Z_{S,1}$ and S is as stated in the main text.

The probability that $S=i$ and $Z_{S,1}$ exceeds c is equal to

$$\begin{aligned}
&\int_c^\infty f_{Z_{S,1},S}(z,i)\,\mathrm{d}z \\
&\quad = \int_{z=c}^\infty \int_{x=-\infty}^\infty \frac{1}{\sqrt{(V_1/2)}}\phi\left(\frac{x-z}{\sqrt{(V_1/2)}}\right) f_{X_S,S}(x,i)\,\mathrm{d}x\,\mathrm{d}z \\
&\quad = \int_{x=-\infty}^\infty f_{X_S,S}(x,i) \int_{z=c}^\infty \frac{1}{\sqrt{(V_1/2)}}\phi\left(\frac{x-z}{\sqrt{(V_1/2)}}\right)\,\mathrm{d}z\,\mathrm{d}x \\
&\quad = \int_{-\infty}^\infty f_{X_S,S}(x,i)\left[1-\Phi\left(\frac{c-x}{\sqrt{(V_1/2)}}\right)\right]\mathrm{d}x
\end{aligned}$$

as stated.

ACKNOWLEDGEMENTS

The authors are grateful to John Whitehead for helpful discussions, to Matthew Murdoch for computational assistance, to Janssen for permission to use the data in Section 3 and to two referees for their comments on earlier versions of the paper.

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